

A. Configuration-Route

DUT issues configuration write



FPGA emits ordered control event



Host software back-end updates image and affected BAR route

happens-before



Shadow update and matching ACK



DUT write completes

B. Submission-DMA

DUT writes queue entry then rings doorbell



FPGA accepts BAR event



Device-semantic adapter observes entry and translates DMA references

happens-before



Translated command becomes visible



Physical doorbell is forwarded

C. Data-Completion

Physical device completes DMA



Payload and writebacks become visible in DUT memory

happens-before



Device-semantic adapter publishes driver-visible completion state



Pending state is updated



Virtual INTx level may change